

Secular Morality and Data Ethics: The Need for a Cohesive Moral Framework

In June of 2023, I penned an essay, “What Hath Tech Wrought?” which argued GPT AI models were being wielded nihilistically and absent consumer protections, GPT AI posed a threat to the moral fabric of society, employment, and individual safety. Unfortunately, my essay’s proximity to the release of GPT AI rendered it more observational than prescient. Two years later, a course in data ethics has exposed me to readings from *97 Things About Ethics Everyone in Data Science Should Know* (Franks, 2020), as well as the books *More Than A Glitch* (Broussard, 2023), *Weapons of Math Destruction* (O’Neil, 2016), and *The Anxious Generation* (Haidt, 2024), and it occurred to me that I was getting a formulation of when algorithms were harmful, how that harm manifested, and what ought be done. Essentially, I was getting the solution, not the calculus. This opened my eyes to a limitation of my 2023 essay: it assumed a shared moral framework that may not exist. My goal here is to make the case that Immanuel Kant’s deontology, modified to include both *Care Ethics* and *Justice Ethics*, is a path out of nihilism that does not require an embrace of religion, but an embrace of reason, logic, and psychology to preserve human dignity. This framework should provide data scientists with the tools they need to create ethical algorithms.

In 1882, Friedrich Nietzsche published *The Gay Science*, which included “The Parable of the Madman,” a story in which Nietzsche declared God was dead (Nietzsche, 2006). Nietzsche’s claim was anything but a salubrious shaking-off of the shackles of morality that limited hedonistic pursuits. It was a lament. The Madman, looking for God, was mocked by a crowd of onlookers for his pursuit of the lost deity. “Where is God?” he asked and proceeded to answer with, “I mean to tell you. *We have killed him* – you and I” (Nietzsche, 2006, p. 90). In a metaphor for detachment from a moral framework, Nietzsche wrote, “What did we do when we loosened this earth from its sun?” and asked “Is not the magnitude of this deed too great for us? Shall we ourselves have to become Gods, merely to seem worthy of it?” (Nietzsche, 2006, p. 90-1). Effectively, absent the moral ethos provided by the God of Judaism and Christianity, humans are as the Earth untethered from the sun, adrift in the vastness of space, and must become omniscient as God to divine a moral framework suitable for all.

Americans are increasingly burdened by such a task. In 2024, the Pew Research Center published polling data that showed 28% of U.S. adults were religiously unaffiliated (Smith et al., 2024, p. 6). Of this population, 82% reported “logic and reason” were pathways to morality (Smith et al., 2024, p. 15). What is more, 6% of Americans were affiliated with non-Christian religions (Smith, 2021, p. 4). Because of the diverse populations in the United States, and the separation of church and state as articulated in the First Amendment, Americans should find it neither feasible nor desirable to impose any religion as a practical solution for a moral framework through which one can act ethically when engaged in data science. Perhaps the religiously unaffiliated were on to something: the pathway to morality is logic and reason.

Morality, according to Immanuel Kant, is the product of reason. In *Groundwork of the Metaphysics of Morals*, Kant defines morality as “the relation of actions to the autonomy of the will, that is, to a possible giving of universal law through its maxims” and that humans are “rational beings” elevated by reason so as to be “givers of universal law” (Kant, 1997, p. 80, 84). Rational beings formulate universal law through the “categorical imperative” that mandates they “*act in accordance with a maxim that can at the same time make itself a universal law*” (Kant, 1997, p. 44). Kant clarifies that a rational being not only “gives universal laws ... but is also himself subject to these laws” (1997, p. 41). Thus, a collective of rational beings belongs to what Kant calls “*a kingdom of ends*” which he defines as “a systematic union of various rational beings” operating in

a system of “a whole of all ends in systematic connection (a whole both of rational beings as ends in themselves and of the ends of his own that each may set himself)” (1997, p. 41).

In Kant’s kingdom of ends, it is dignity that elevates humanity. Kant explains that “everything either has a price or a dignity” and that which “has a price can be replaced by something else as its *equivalent*; what on the other hand is raised above all price and therefore admits no equivalent has dignity” (1997, p. 42). Kant concludes that anything that can be purchased is simply a means to experience that “purposeless play of our mental powers” but that which is an “end in itself has not merely ... a price, but an inner worth, that is, *dignity*” (1997, p. 42). Reason enables humans to create universal laws which treat themselves and others as ends, thus they are rational beings. Because rational beings only impose laws on others, they themselves must follow, they have inherent dignity. In fact, humanity, says Kant, “is that which alone has dignity” (1997, p. 42). For Kant, “the worth of such a cast of mind,” the rational being who operates under the categorical imperative, is “cognized as dignity and puts it infinitely above all price, with which it cannot be brought into comparison or competition at all without ... assaulting its holiness” (1997, p. 43).

Before proceeding with why deontology is the correct framework, it is necessary to address utilitarianism and post-modern thinking. Utilitarianism is a moral philosophy that argues an action is right if it promotes happiness or pleasure (e.g., good) and wrong if it produces unhappiness (West & Britannica Editors, 2025). In practice, the utilitarian is required to assess the consequence of an action and select the action that produces the maximum good. The founder of utilitarianism, Jeremy Bentham, differed from Kant on reason, arguing “*The question is not, Can they reason? Nor, Can they talk? But, Can they suffer?*” (as cited in Richards, 2015). It is worth noting the “they” to whom Bentham referred was animals, but his comment about suffering serves as a marker for when someone or some thing deserves consideration for the consequence of an action.

Post-modernism, by contrast, is more abstract. The post-modernist rejects objective reality, asserting there is no scientific or objective truth; argues that language itself has no external referent; as such, there is no path to concrete knowledge; thus, it concludes that all theories of reality are equally true because all are equally illegitimate (Dunigan & Britannica Editors, 2025). The formulation that language has no objective meaning is credited to Jacques Derrida (Dunigan & Britannica Editors, 2025), a French philosopher famous for the rejection of Western binaries and hierarchies, who taught at Yale and University of California at Irvine (Britannica Editors, 2025).

These philosophies work as thought experiments but are unsuitable in practice. Post-modernism falls apart precisely because it rejects objectivity. In *My Philosophical Development*, the British philosopher Bertrand Russell (1959) argued that linguistic meaning is acquired through “two habits” he called “active” and “passive” understanding, which leverage association (e.g., connecting a word to its target) and prediction (e.g., hearing a word and expecting the target) to connect language to an external referent, thus establishing “object-words” (p. 146). Russell argued that words must have objective meaning attached to them, even if those words differed between people, because absent objective meaning, words fail to perform their function: communicate observable or internal phenomena. For a human, language learning is intuitive. For machines, it is algorithmic.

Large language models (LLM) are trained in exactly this manner. LLMs are trained through supervised learning: labeled data is fed into the model (e.g., of an image: dog = true versus

dog = false) providing the model with “active learning” and after a series of trials, the LLM is given unlabeled data containing a mix of true targets (e.g., image of a dog) and false targets (e.g., some other image) and tasked with demonstrating “passive understanding” or prediction. If, as the post-modernist suggests, language has no meaning, the machine could make any prediction – that a chair is a dog or a wastebasket is a dog – and it would be equally correct. Were it a human, the LLM would be diagnosed with Broca’s Aphasia, which is characterized by having a correct understanding (e.g., a chair is a chair) but an inability to name objects, repeat phrases, read or write (Cleveland Clinic, 2024). This is at odds with post-modernism because even it requires words with objective meaning to argue that all meta-narratives are equally valid. Therefore, the post-modernist formulation that language has no objective meaning is also incompatible with reality.

But post-modernism is worse than that: it is a construction of absurd academics. In addition to Derrida, other famous post modernists include Michel Foucault, a Marxist philosopher who taught at the University of Paris and University of California at Berkley, and is famous for the concept of power-knowledge, which posits knowledge is an expression of power, not truth (Faubion & Britannica Editors, 2025). There is also Julia Kristeva, another French philosopher and Marxist-educated psychoanalyst who critiqued linguistic formulations as paternal at the University of Paris (Oliver & Britannica Editors, 2025). For various reasons, each of these post-modernist academics has argued language has no objective meaning. “Philosophers and bookish people ... forget that it is the essential function of words to have a connection ... with facts,” wrote Russell, before observing that, “Some modern philosophers ... say that words should never be confronted with facts but should live in a pure, autonomous world where they are compared only with other words” (1959, p. 148). Russell’s formulation of these philosophers and bookish people describe the post-modernist. He concluded that the detachment of words from facts is “one of those views which [is] so absurd that only very learned men could possibly adopt [it]” (p. 148).

If the philosophical take is not compelling, perhaps the psychological take will suffice. Psychologists Paul Ekman and Wallace Friesen (1971) are famous for their observation that there are six foundational emotions shared across cultures (e.g., happiness, sadness, anger, fear, surprise, disgust). *Glæde*, *glück*, *felicità*, *kōfuku*, *счастье*, and *felicidad* are how you say “happiness” in Danish, German, Italian, Japanese, Russian, and Spanish, respectively. These are but a sample of different words with objective meanings associated with but one of six commonly expressed human emotions. “A rose by any other name,” wrote Shakespear in *Romeo and Juliet*, “would smell as sweet.” So, a smile in German, Japanese, Spanish, English – or any other language – is by any other word, a sign of happiness. Even an infant can recognize the difference between a still face and a smiling, attentive one (Tronick et al., 1978). It is deeply unfortunate that post-modernists would not ascribe happiness to a smile because it has many different names. In their bookishness and their philosophizing, they have lost their humanity and peddle their pernicious detachment from reality as intellectual.

Deontology is more than academic blatherskite; it is psychologically protective. Studies on the use of deontological reasoning in moral judgements have shown that refusal to cause harm to maximize outcomes is associated with warmth and morality but less competent when compared to utilitarianism (Rom et al., 2017). Researchers have further found that people who make deontological judgments are “preferred as social partners, perceived as more moral and trustworthy, and are trusted more in economic games” (Everett et al., 2016, p. 772). It is reasonable to assume that these observations stem from Kant’s insistence that rational beings possessing dignity should never be treated as a means, but as an end.

Indeed, life would be difficult to live if people had to constantly evaluate every moment of their life as possibly conflicting with the utilitarian's concept of maximum good. In an extreme example, Russian socialists in the 20th century concluded that, for the good of the many, Ukrainian agricultural production in its entirety should be redirected to Russia proper (Asaulyuk, 2021). From 1932 to 1933, millions of Ukrainians died from starvation, and as the bodies began to amass, Soviet propagandists produced posters kindly reminding the starving masses "It's wrong to eat your children" (Asaulyuk, 2021). Though utilitarianism may be received as more competent, the Ukrainian Genocide is a chilling reminder of just how cold utilitarianism can be.

Despite being perceived as a preferred moral framework, deontology has been critiqued as male centric. In the field of psychology, Lawrence Kohlberg is best known for his formulation of moral development, the highest stage of which was universal principle (McLeod & Guy-Evans, 2025). In Kohlberg's model, universal principles were arrived at through Kant's categorical imperative. Harvard psychologist Carol Gilligan (1993) famously criticized Kohlberg's stages of moral development in her book, *In A Different Voice*. Gilligan argued that Kohlberg's highest stage of moral development – universal principles – largely precluded women, whose moral reasoning tended to consider relational contexts and the needs of others, in what she called an Ethic of Care, as compared to an Ethic of Justice. In Jonathan Haidt's (2024) book, *The Anxious Generation*, the dichotomy of the male Ethic of Justice and female Ethic of Care is described as "agency" and "communion" where agency was characterized as a tendency toward individualism, competency, efficiency, and assertiveness, while communion was characterized as a tendency toward integration into social units and caring for others and cooperation (p. 152). Thus, to divine a proper moral framework, it is necessary to nest Gilligan's Ethic of Care and Kohlberg's Ethic of Justice within deontology as dual expressions of dignity.

Dignity must be conceived of as that which preserves order and individuation while at the same time, facilitating relationship formation. Beginning in childhood, boys and girls express agency and commune in the way they play. Writing for *Social Problems*, researcher Janet Lever (1976) observed that boys prefer to play competitive games in parallel, while girls prefer cooperative turn-taking games; that boys emphasize rules and structure, while girls will relax rules in favor of inclusivity; and games that boys play last longer because when rules are broken, boys get as much satisfaction arguing over rules as they do playing the game; whereas girls will stop the game to prevent relational conflicts.

Combining Lever's research with Kohlberg, Gilligan, and Kant, dignity must be that which both preserves a system of rules and individuation where males can compete in accordance with those rules to demonstrate competence such that a hierarchy of competence is established. At the same time, a system must also not become so encumbered by rules and structure that it arbitrarily excludes people from participation, thus preventing relationship formation. Dignity, then, is that which promotes justice and modifies the rules for the sake of care.

Having addressed gender differences in moral development, it becomes necessary to address dignity in terms of cognitive development. The nexus of Kant's deontology is reason. Problematically, human capacity for reason develops later in life. Writing for *Neuropsychiatric Disease and Treatment*, Arain et al. (2013) note that differential development of the brain from birth to about 25 years of age affects an individual's capacity to regulate emotions and make decisions. The two brain systems they focus on are the limbic system and the prefrontal cortex. The limbic system includes the brain's reward center (e.g., nucleus accumbens) which is associated

with impulsive and risky decision making, and the brain's emotion center (e.g., amygdala) which is associated with emotional volatility and sex drive, resulting in poor self-regulation.

In contrast to the limbic system, the prefrontal cortex is associated with behavioral regulation (e.g., executive function), which includes impulse control, emotional regulation, organizing thoughts, problem solving, and abstraction (e.g., considering consequences of one's actions). Critically, the limbic system fully develops between 10-13 years of age, while the prefrontal cortex develops between 22-25 years of age (Kassell & Gill, 2023). This evidence suggests that reason is impaired from birth through adolescence and even into early adulthood. Thus, dignity is something people develop into and must be protected during development, when children are least capable of maintaining their dignity.

This updated version of deontology is now adequate as a framework for evaluating data ethics. Beginning with the categorical imperative, an algorithm must be designed such that its creator is at the same time subject to the analysis and disclosure rules of the algorithm. No algorithm shall be implemented that disrupts the healthy formation of hierarchical systems of competency nor disrupts an individual's ability to form healthy relationships (e.g., undermines interpersonal relationships or is arbitrarily / tyrannically exclusive). "Healthy" should be taken to mean systems that facilitate prosocial behavior. No algorithm or system of data collection is ethical that takes advantage of an individual whose capacity for reason is developing. Finally, no algorithm or data collection system is ethical if it diminishes human dignity. This is by no means a complete list, and it may not even be a properly defined list. Rather, it is meant to be the starting point in a long continuum of data ethics. The heretofore discussed deontological framework should guide this continuum.

To come this far and not conclude with some examples of unethical algorithms would be malpractice. Perhaps the most pernicious algorithms of the modern age are those that take advantage of the human tendency to attend to moral content. This harms those who are chronically online, beginning with the most vulnerable: teenagers. On average, 51% of teens aged 13 to 19 have been found to spend about 4.8 hours per day on social media (Rothwell, 2023; Zablotzky et al., 2024). Problematically, social media companies (e.g., YouTube, Twitter, Facebook, Reddit) use algorithms that take advantage of our tendency to attend to moral content, leading to maladaptive behavior (Brady et al., 2023). One study suggested that humans evolved to evaluate moral content in small groups, and that online interactions via social media flooded users with moral content, and contributed to an evolutionary mismatch (Li et al., 2018).

Constant evaluation of moralized content and the extent to which teens spend more time online is potentially concerning as it may impact brain development. Moral content often involves emotionally valenced content (Brady et al., 2023). Given the differential development of the limbic system, which processes and interprets fear, pleasure, and sadness (Rajmohan & Mohandas, 2007), and the prefrontal cortex, which regulates responses of the limbic system (Hathaway & Newton, 2025), it is reasonable to conclude that teens are uniquely susceptible to the effects of processing moral content.

Researchers De et al. (2025) examined the relationship between social media usage and neural development in teens with social media addictions. Their research revealed teenagers addicted to social media (i.e., an uncontrollable urge to browse social media that interferes with daily life, including anxiety, depression, and diminished mental health) demonstrated low levels of activity in the prefrontal cortex (e.g., ventral medial and medial), posterior cingulate cortex, and

right inferior frontal gyrus. These differences were associated with addiction to social media two years later, and symptoms of anxiety and depression. Taken together, social media algorithms that over stimulate users with moral content, especially teenagers, are unethical because they not only cause emotional harm, but they inhibit the development of brain systems associated with reason; therefore, they diminish the dignity of their users.

For the secularist, “God is dead. God remains dead. And we have killed him,” wrote Nietzsche (2006), “whoever is born after us – for the sake of this deed he will belong to a higher history than all history hitherto,” he concluded (p. 90-1). Absent God, and absent the ability to enforce a national religion, and in the face of an increasingly secular population, it becomes necessary to turn to reason and logic. Post-modernism and utilitarianism fall short of carrying humanity to a higher history. In the case of the Ukrainian Genocide, utilitarianism was used to justify the deaths of millions. Post-modernism detaches language from its meaning and denies objectivity, putting it at odds with not only basic machine learning, but humanity itself. To meet Nietzsche’s charge of being born into a higher history, people need something more than the intellectual blatherskite offered by post-modernists and the all-too-flexible utilitarianism.

Immanuel Kant’s deontology, by contrast, provides a sufficient framework for secular morality. But deontology is not without its criticism, namely its male-centric nature. By incorporating *Care Ethics* and *Justice Ethics* into Kant’s formulation of dignity, both sexes are represented. By acknowledging dignity exists on a continuum, deontology becomes robust enough to protect those most vulnerable to algorithms that prey on the pre-rational being: children. To be clear, this is a calculus, not a solution. It is a tool for data scientists to evaluate whether their algorithms treat users as a means or an end. More importantly, it is a pathway out of nihilism that requires no religious affiliation, which is suitable for all Americans.

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